GREEN SPACE CONFIGURATION AND PLUVIAL FLOOD RISK

Group C

ARFW0501 Applied Spatial Analysis for Sustainable Urban Development

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RESEARCH FOCUS

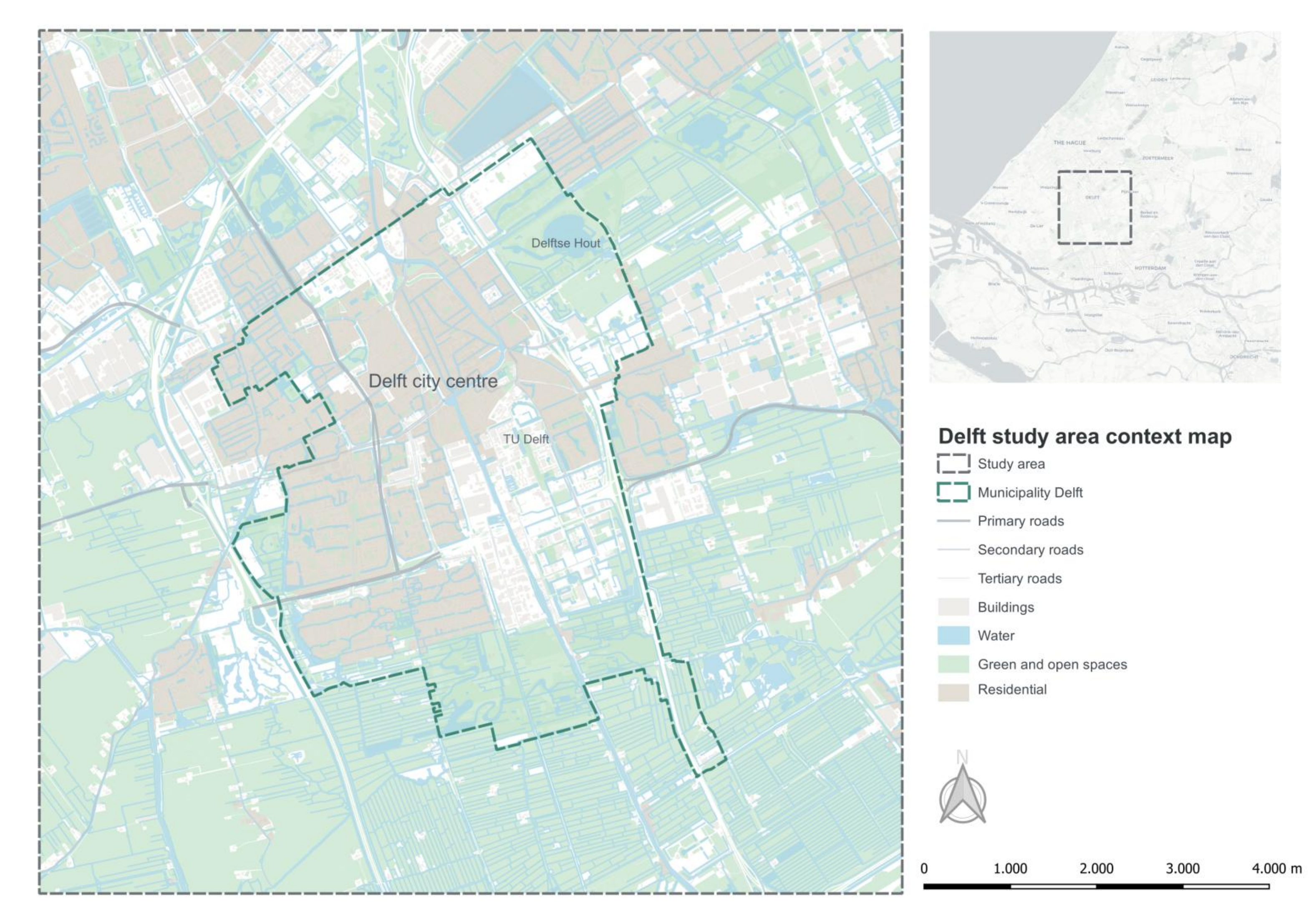
THEME: ECOLOGY + FLOOD

CITIES: DELFT AND XI'AN

MAIN RESEARCH QUESTION: HOW DOES THE SPATIAL CONFIGURATION OF GREEN SPACES INFLUENCE PLUVIAL FLOODING IN URBAN AREAS SUCH AS DELFT AND XI'AN?'

STUDY AREA DELFT

- 10 × 10 KM STUDY AREA FULLY CONTAINS DELFT
- MUNICIPALITY BOUNDARY INCLUDED FOR SPATIAL REFERENCE
- SURROUNDING CANALS, GREEN AREAS AND URBAN EDGES PROVIDE CONTEXT

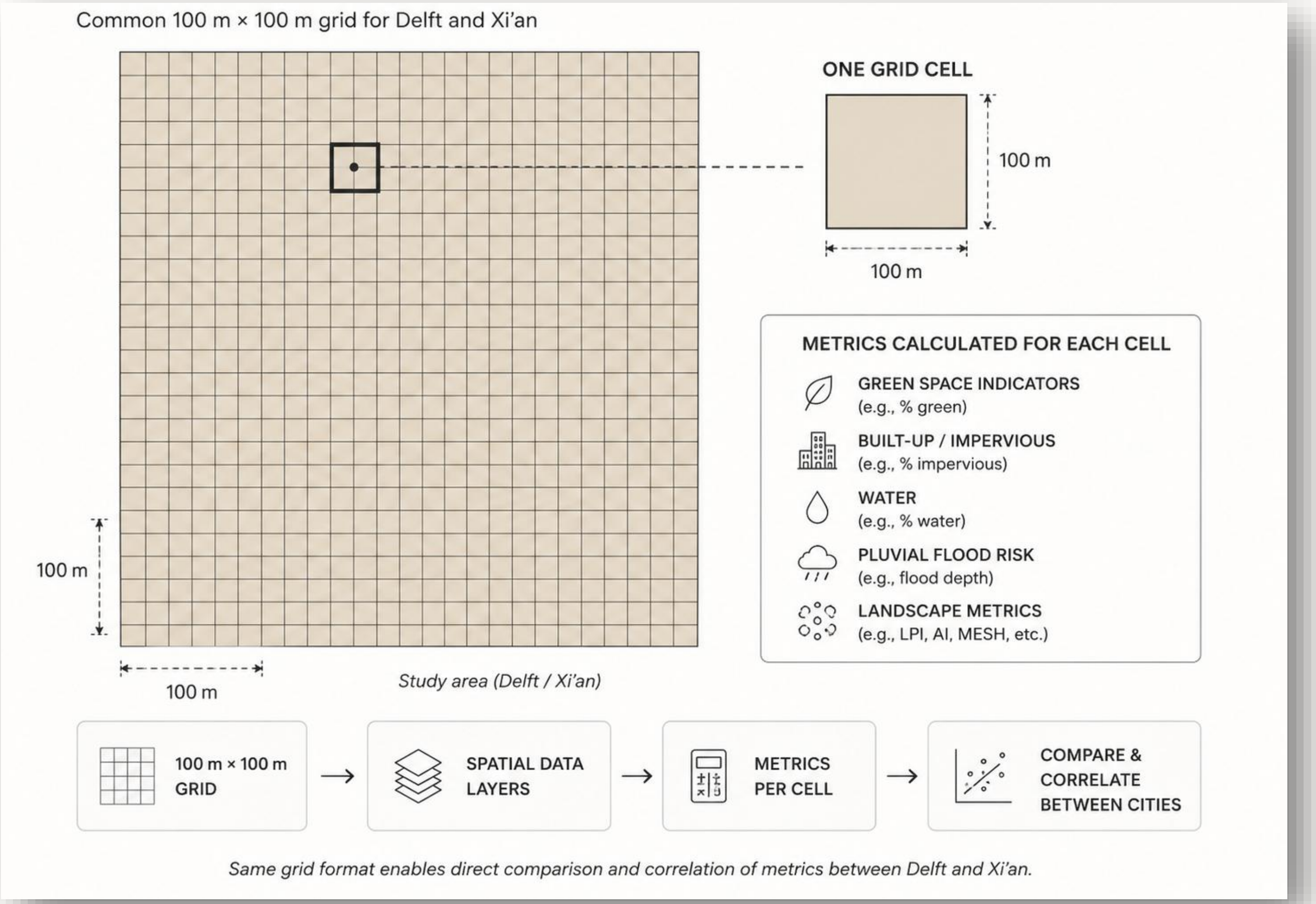


STUDY AREA XI’AN

- XI’AN IS MUCH LARGER THAN DELFT
- SAME 10 × 10 KM GRID CAPTURES A CENTRAL URBAN SUBSET
- AREA INCLUDES THE HISTORIC CITY CENTRE, CITY WALL AND MAJOR GREEN HERITAGE PARKS

METHODOLOGY

- CREATE A COMMON 100 × 100 M GRID
- CALCULATE GREEN SPACE CONFIGURATION METRICS
- DERIVE FWEI-BASED FLOOD INDICATORS
- CREATE GREEN SPACE AND FLOOD PATTERN TYPES
- COMPARE TYPOLOGIES, CORRELATIONS AND DEM VARIABLES



GREEN SPACE METRICS

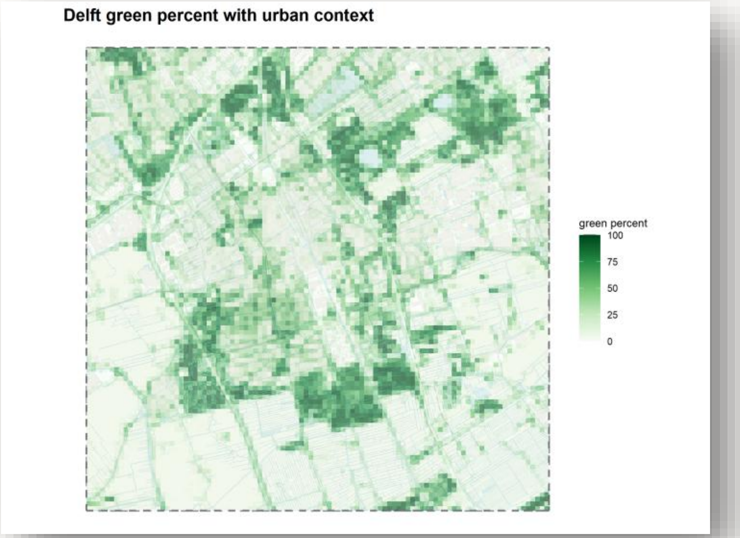
- **GREEN SPACE AMOUNT:** HOW MUCH OF EACH CELL IS GREEN **(GREEN_PERCENT**

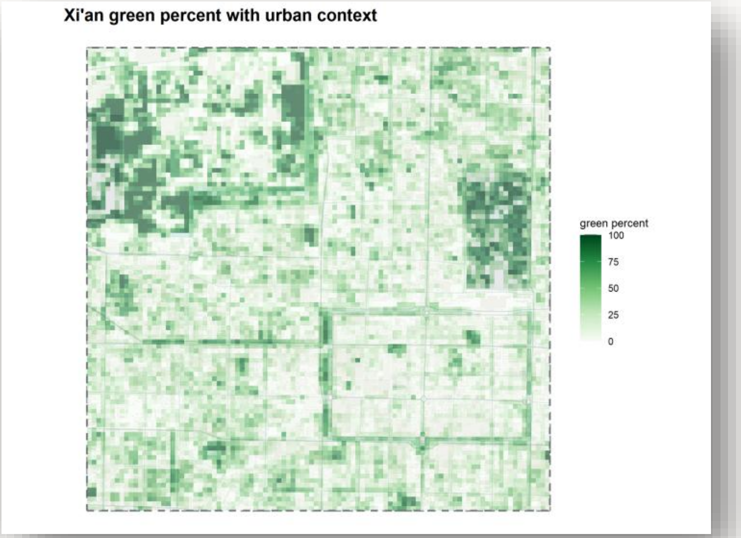
- **PATCH SIZE:** WHETHER GREEN AREAS ARE SMALL OR LARGE **(AREA)**

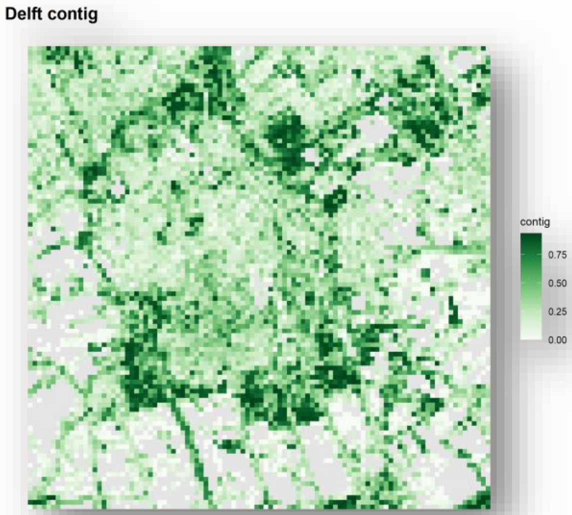
- **PATCH SPREAD:** HOW FAR GREEN PATCHES EXTEND SPATIALLY **(GYRATE)**

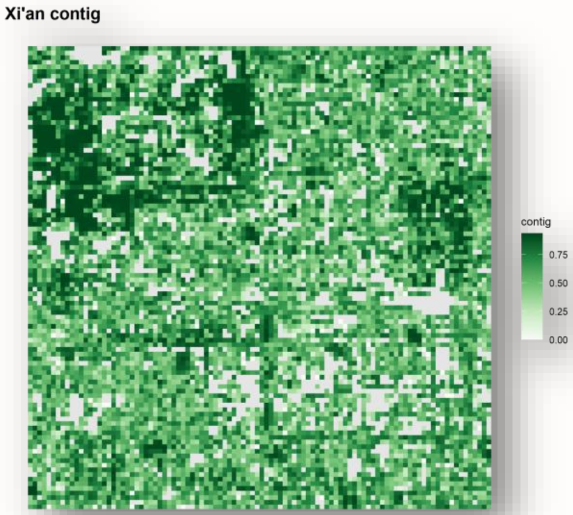
- **INTERNAL CONNECTIVITY:** WHETHER GREEN PATCHES ARE COMPACT AND CONNECTED **(CONTIG)**

- **PATCH ISOLATION:** DISTANCE TO THE NEAREST OTHER GREEN PATCH **(ENN)**



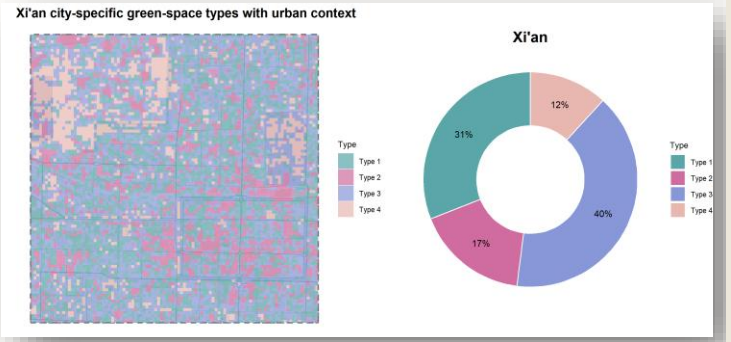


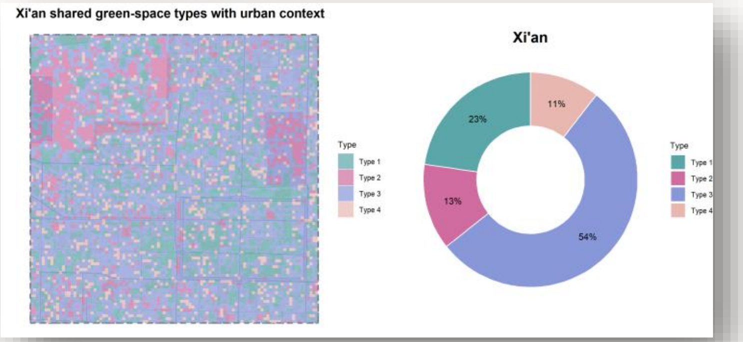


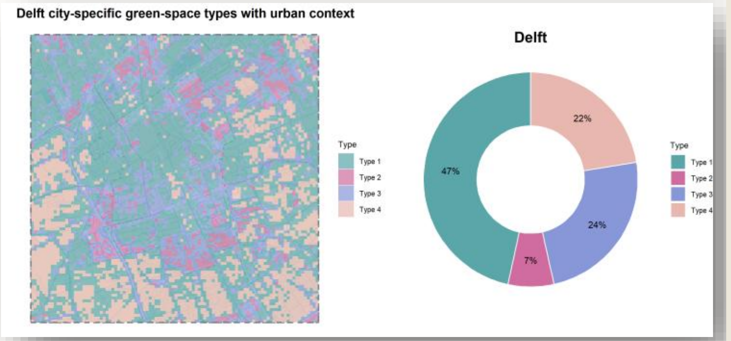


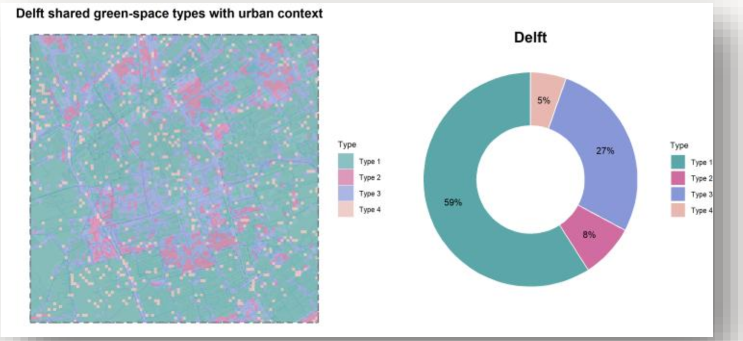
GREEN SPACE TYPOLOGIES

- K-MEANS CLUSTERING BASED ON GREEN METRICS
- 4 GREEN SPACE TYPES
- SHARED TYPOLOGIES USED FOR DELFT–XI’AN COMPARISON
- SEPARATE TYPOLOGIES USED AS CITY SPECIFIC CHECKS







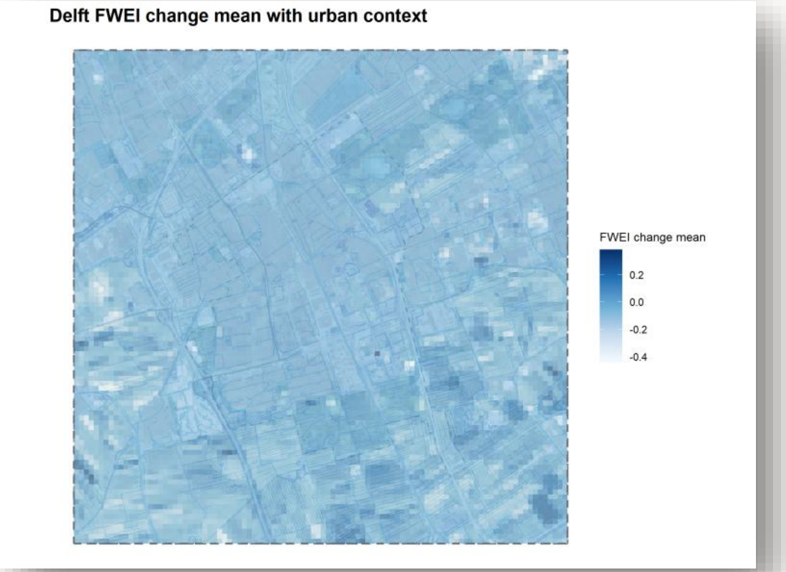


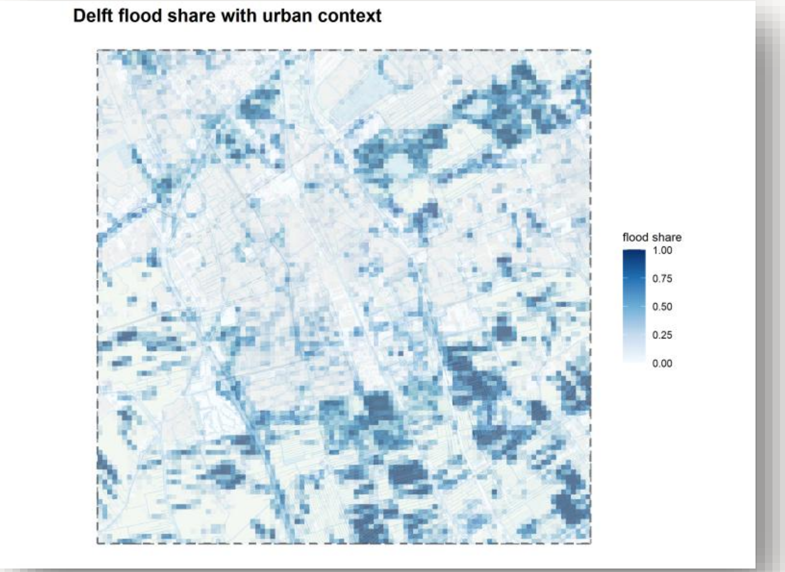
SEPERATE

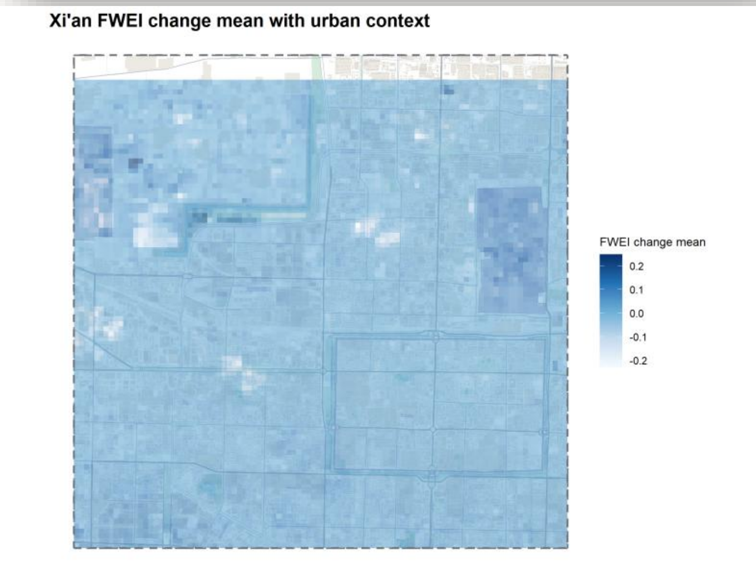
SHARED

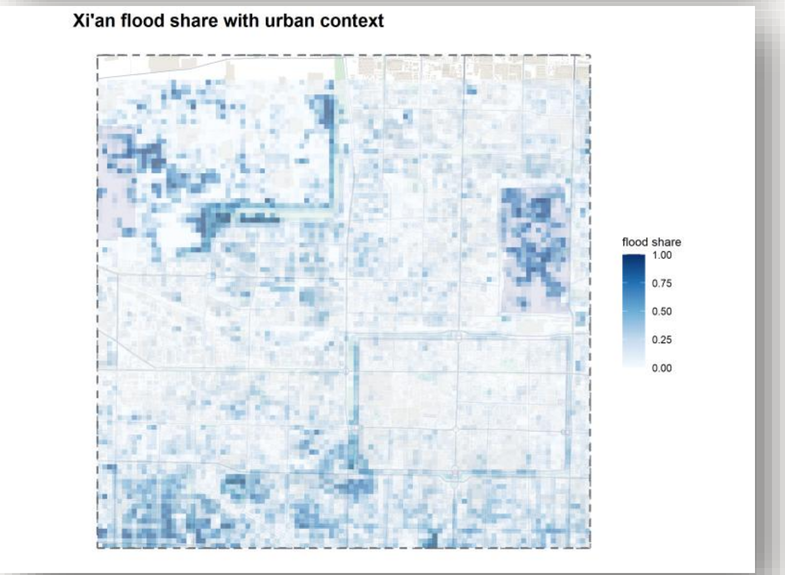
FWEI-BASED FLOOD INDICATORS

- FWEI CHANGE = AFTER RAINFALL – BEFORE RAINFALL
- FLOOD SHARE = PROPORTION OF DETECTED SURFACE-WATER INCREASE
- FLOOD-PATTERN METRICS DESCRIBE CLUSTERING AND FRAGMENTATION

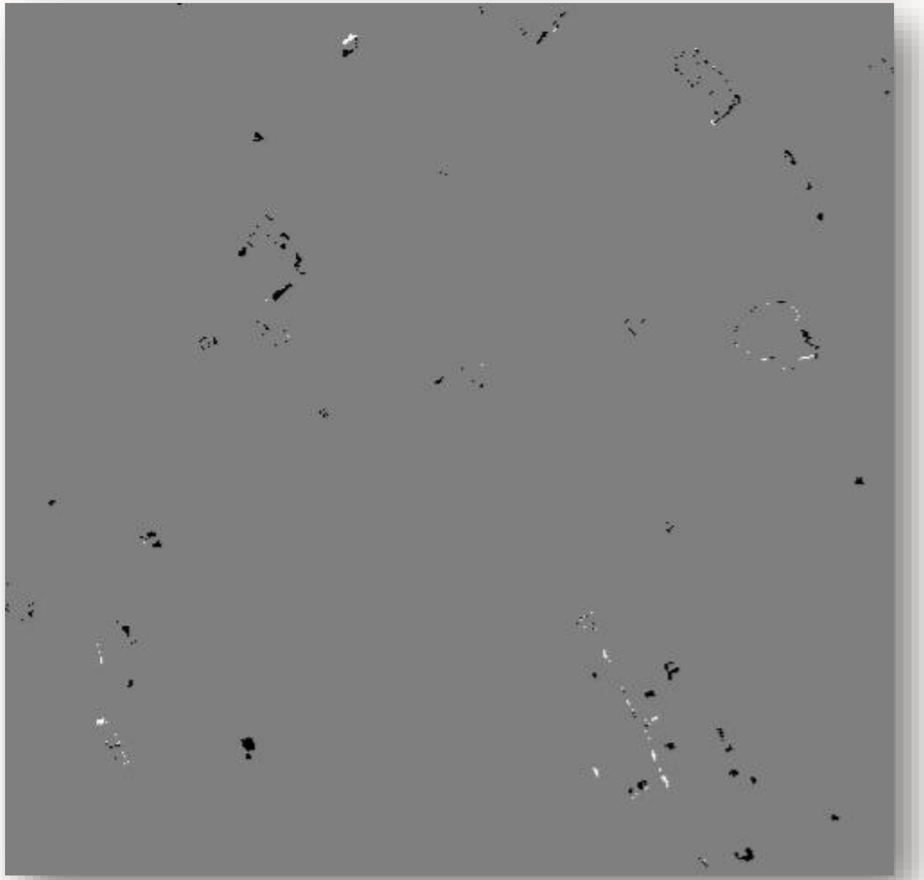




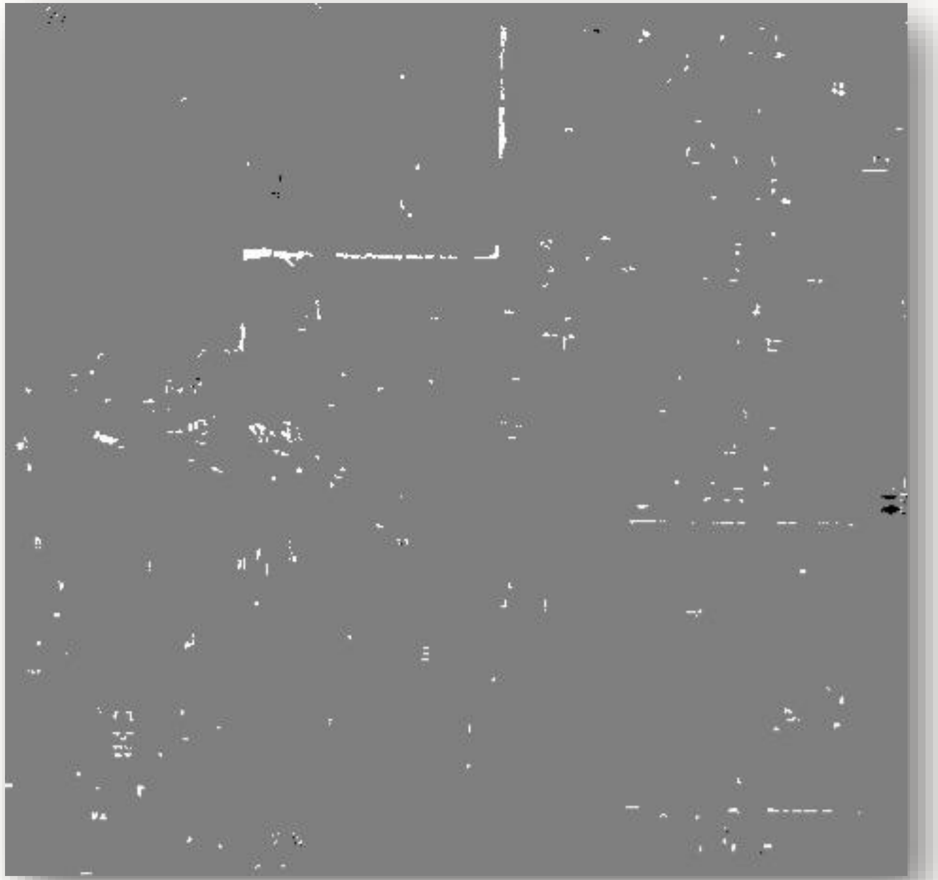




FWEI DERIVED FLOOD MASK



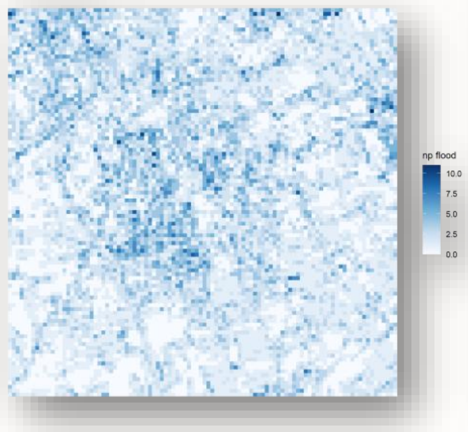
Delft flood mask layer



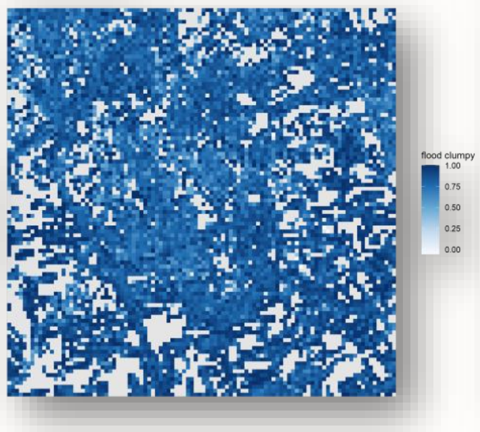
Xian flood mask layer

FLOOD METRICS

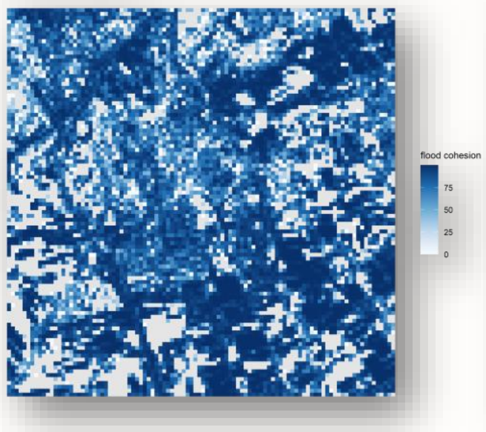
Delft np flood



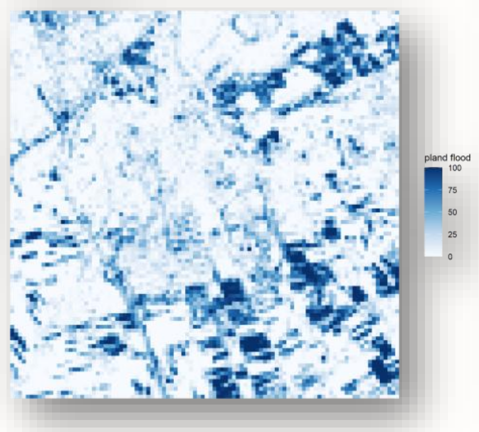
Delft flood clumpy



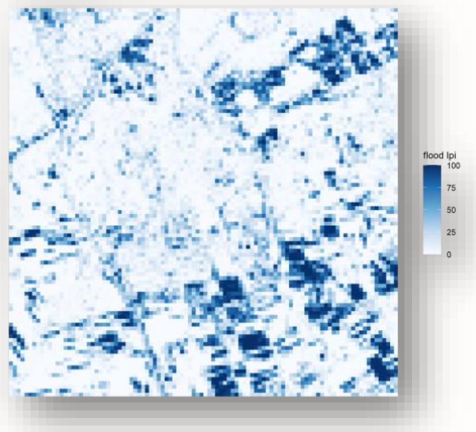
Delft flood cohesion



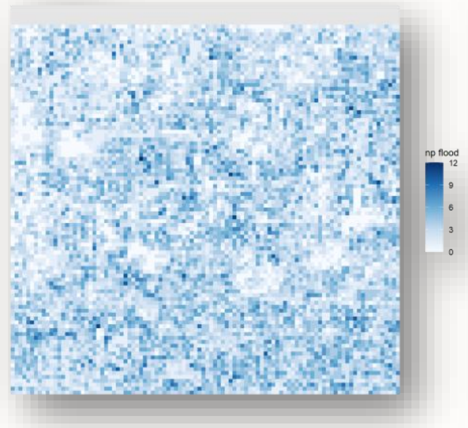
Delft pland flood



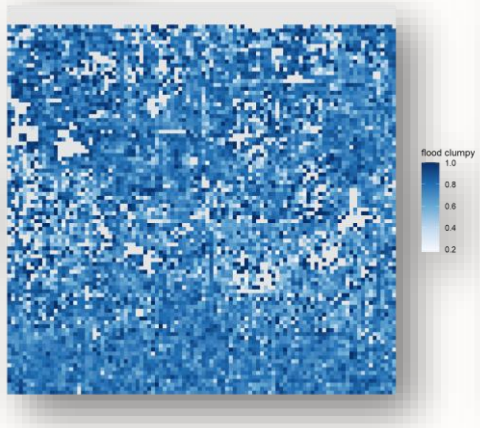
Delft flood lpi



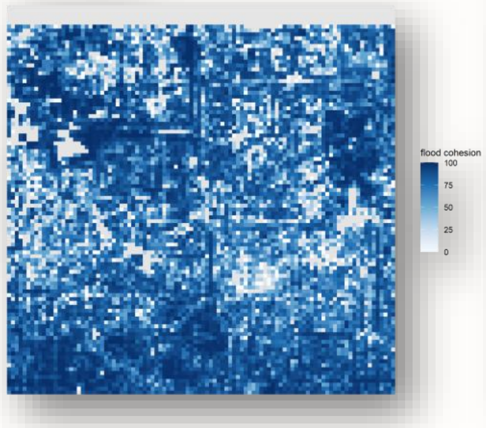
Xi'an np flood



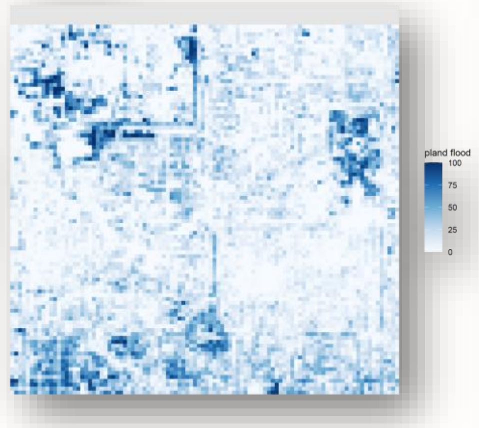
Xi'an flood clumpy



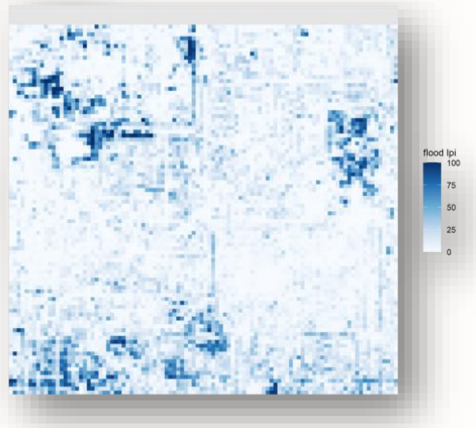
Xi'an flood cohesion



Xi'an pland flood

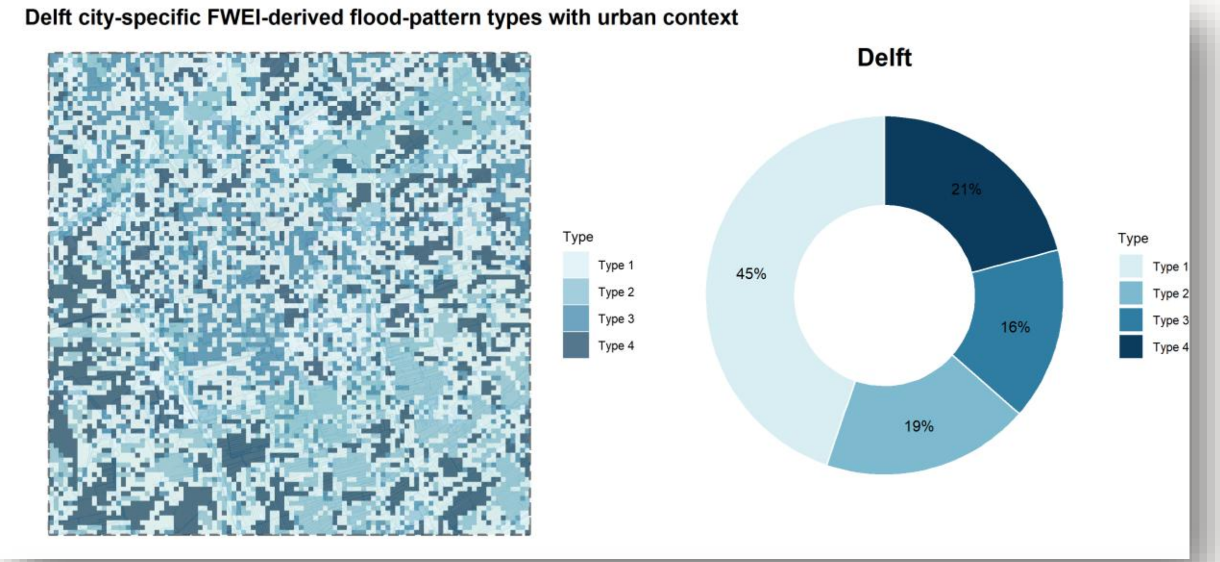


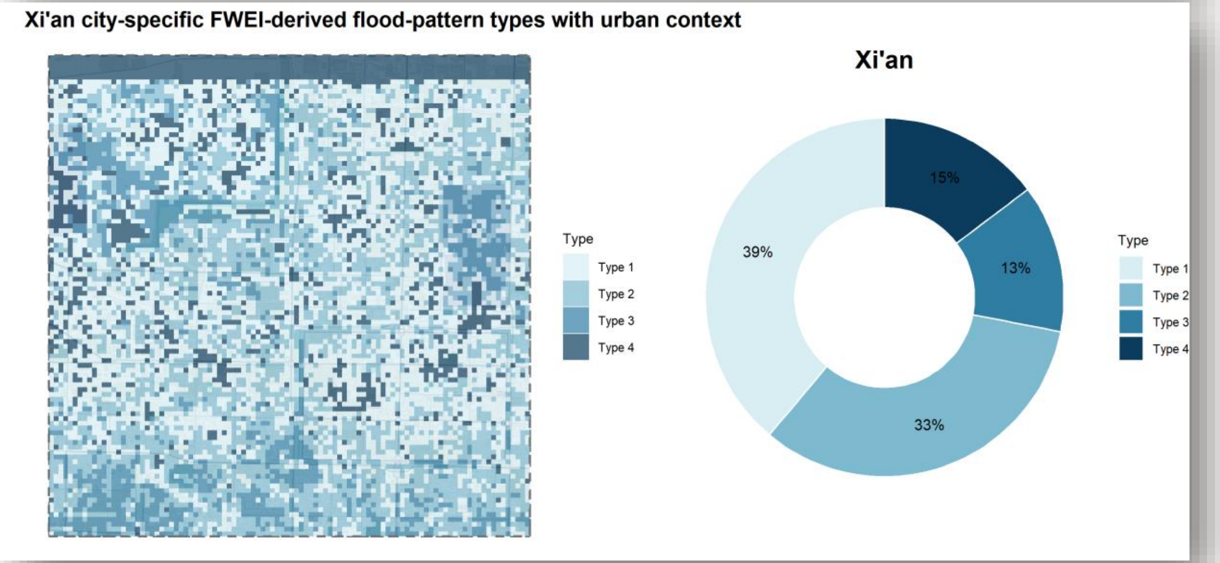
Xi'an flood lpi



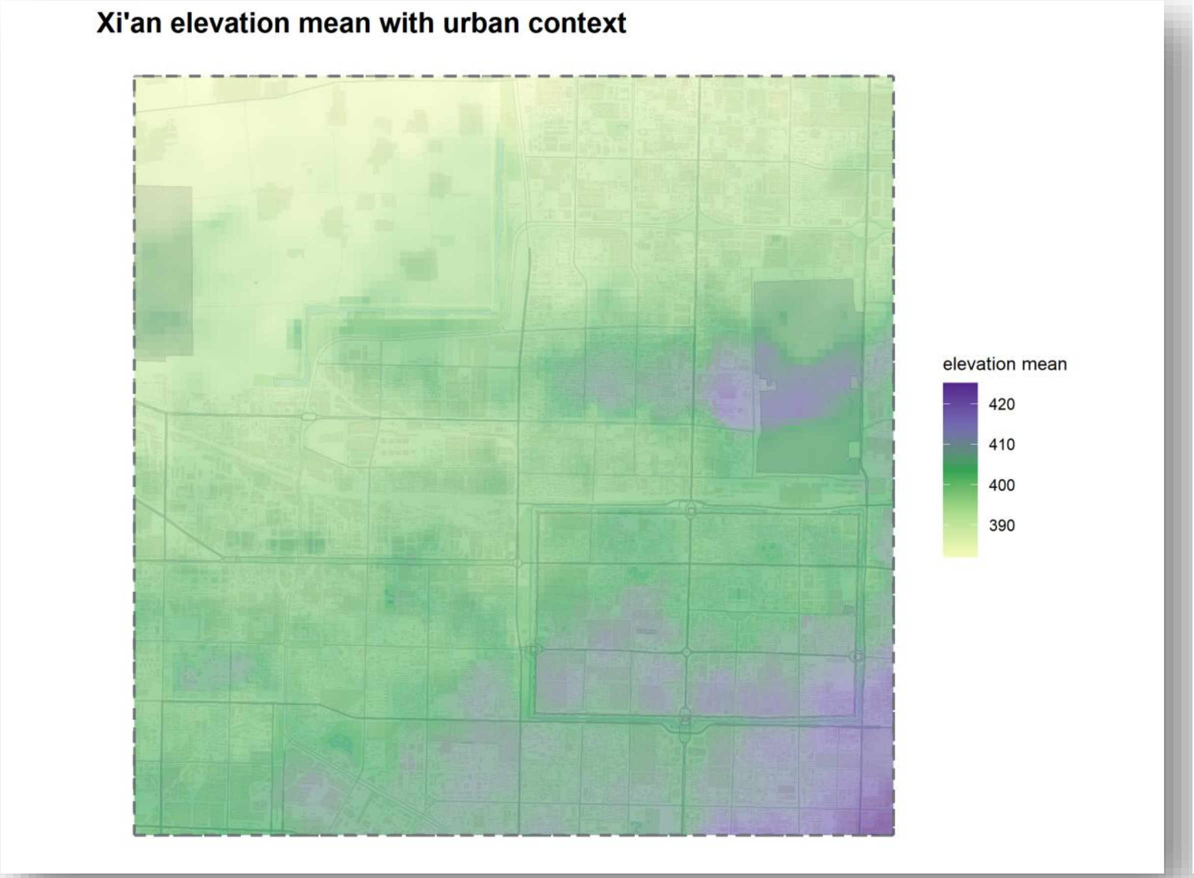
FLOOD PATTERN TYPES

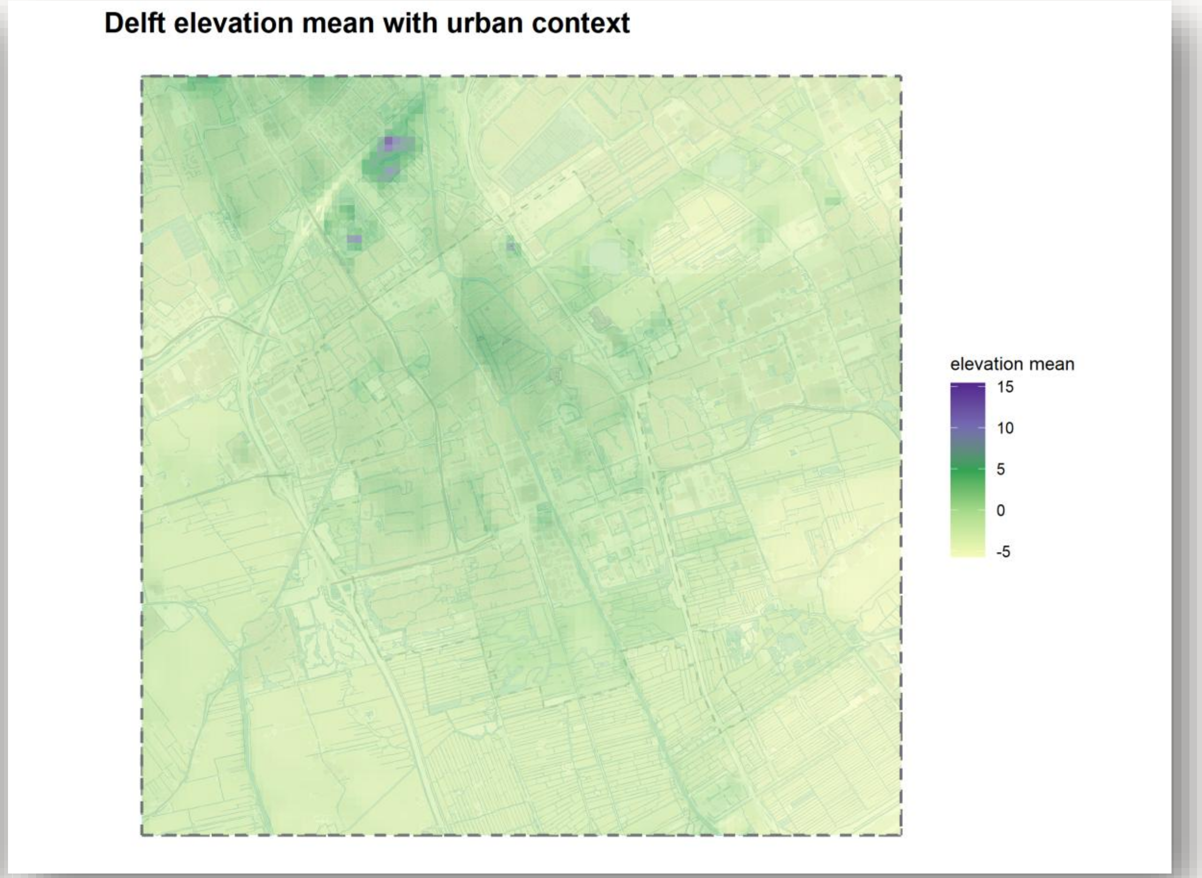
- BASED ONLY ON FWEI-DERIVED FLOOD METRICS
- DESCRIBES FLOOD EXTENT, PATCH NUMBER AND CLUSTERING
- SEPARATE FROM GREEN-SPACE TYPES TO AVOID CIRCULAR REASONING





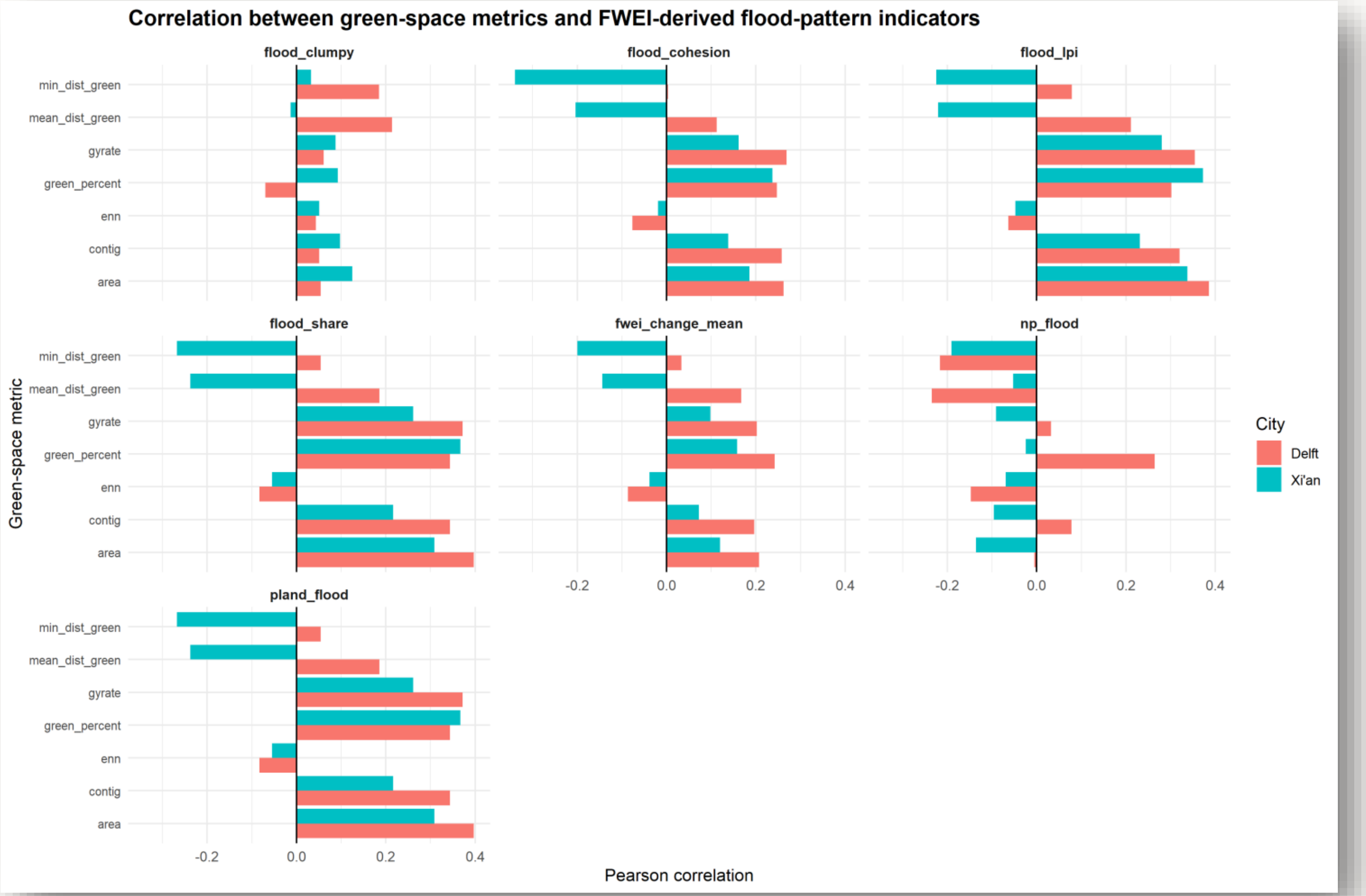
DEM VARIABLES





MAIN RESULTS 1/2

- PEARSON CORRELATION USED
- MORE GREEN DOES NOT ALWAYS MEAN LESS DETECTED WATER
- SOME GREEN AREAS MAY STORE WATER TEMPORARILY



MAIN RESULTS 2/2

Key point	Meaning
Weak overall relationships	Correlations between green configuration and flood metrics are mostly weak, usually below $r = \pm 0.20$.
More significant results in Xi’an	Xi’an has more flooded cells than Delft, 663 vs 129, so it has more statistical power.
Mostly positive correlations	Better green configuration often co-occurs with more detected water, likely due to topography or open-area water storage, not causation.
Delft exception	contig vs np_flood in Delft is negative, $r = -0.200$, meaning compact green is linked to fewer separate flood patches.
Main comparison	Delft suggests some fragmentation-suppression effect, while Xi’an mainly shows co-location between green areas and detected water.

DISCUSSION – PIPELINE REPRODUCIBILITY

Manual pre-processing

Find layers, clip to study area, reproject CRS, organise input files.

Automated analysis

Scripts calculate green, flood, distance-to-green, and DEM metrics automatically.

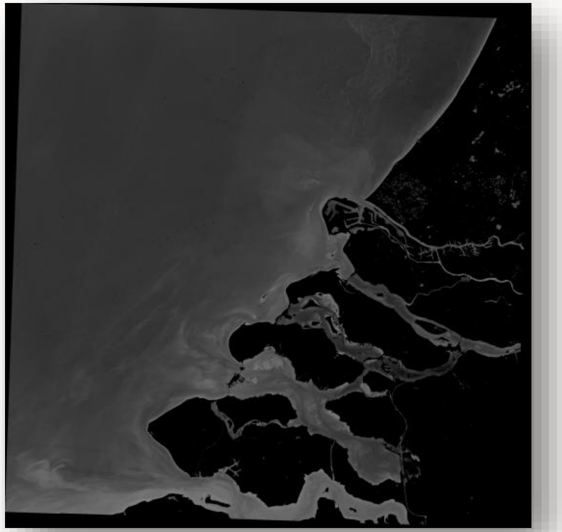
Comparable outputs

Same outputs for both cities: maps, CSVs, typologies, and correlation plots.

LIMITATIONS

- FWEI is a **proxy**, not measured flood depth
- Results show **association**, not causation
- Data **quality** and detectability differ between cities
- **Drainage** and **rainfall** intensity are not fully included

Heavy rainfall dataset Delft



FWEI layer Delft